Task 1: Dot Products and Gradients

1(a) Computing Predictions Using a Linear Model

Given Information

We are given a linear model:

$$\hat{y} = X\theta$$

The input feature matrix is:

$$X = \begin{bmatrix} 1 & 25 & 50 & 2 & 50 \\ 1 & -10 & 50 & 0 & 10 \end{bmatrix}$$

The parameter vector is:

$$\theta = \begin{bmatrix} 0.2 \\ 0.64 \\ 0 \\ 1 \\ 0.14 \end{bmatrix}$$

We need to calculate two predictions, $\hat{y}_1$ and $\hat{y}_2$, by multiplying each row of the input matrix with the parameter vector.

Concept: Dot Product

A dot product is calculated by multiplying corresponding elements of two vectors and adding the resulting values.

For example:

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = ac + bd$$

We apply the same principle to our five input features.

Step 1: Calculate the First Prediction

The first row of the input matrix is:

$$X_1 = \begin{bmatrix} 1 & 25 & 50 & 2 & 50 \end{bmatrix}$$

The prediction is calculated as:

$$\hat{y}_1 = 1(0.2) + 25(0.64) + 50(0) \\ + 2(1) + 50(0.14)$$

Simplifying:

$$\hat{y}_1 = 0.2 + 16 + 0 + 2 + 7$$

Therefore:

$$\boxed{\hat{y}_1 = 25.2}$$

Step 2: Calculate the Second Prediction

The second row of the input matrix is:

$$X_2 = \begin{bmatrix} 1 & -10 & 50 & 0 & 10 \end{bmatrix}$$

The prediction is:

$$\begin{aligned} \hat{y}_2 = 1(0.2) + (-10)(0.64) + 50(0) \\ + 0(1) + 10(0.14) \end{aligned}$$

Simplifying:

$$\hat{y}_2 = 0.2 - 6.4 + 0 + 0 + 1.4$$

Therefore:

$$\boxed{\hat{y}_2 = -4.8}$$

Final Answer for 1(a)

The two predictions are:

$$\boxed{\hat{y} = \begin{bmatrix} 25.2 \\ -4.8 \end{bmatrix}}$$

1(b) Understanding and Calculating Gradients

Background: What Is a Gradient?

A gradient is a mathematical vector containing all the partial derivatives of a function.

It tells us how the function changes when we change its parameters.

In machine learning, our objective is to minimise the error between predicted values and actual values.

We use a technique called **Gradient Descent** to achieve this.

1. Partial Derivative

A partial derivative measures how a function changes when we change one variable while keeping the other variables constant.

For a function with two variables:

$$f(\theta_1, \theta_2)$$

The partial derivatives are:

$$\frac{\partial f}{\partial \theta_1} \qquad \text{and} \qquad \frac{\partial f}{\partial \theta_2}$$

2. Gradient

The gradient combines all the partial derivatives into a single vector:

$$\boxed{\nabla_\theta f = \begin{bmatrix} \frac{\partial f}{\partial \theta_1} \\ \frac{\partial f}{\partial \theta_2} \end{bmatrix}}$$

The gradient points in the direction in which the function increases most rapidly.

To decrease the error, we move in the opposite direction of the gradient.

This process is called Gradient Descent.

Gradient Descent Update Formula

The update formula is:

$$\boxed{\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla_\theta f}$$

Where:

- θ_{old} = Current parameters.
- θ_{new} = Updated parameters.
- η = Learning rate.
- $\nabla_\theta f$ = Gradient.

The learning rate determines the size of each update step.

Given Information

We are given:

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad y = 3$$

The linear model predicts:

$$\hat{y} = \theta^T x$$

Expanding the dot product:

$$\hat{y} = \theta_1(1) + \theta_2(2)$$

Therefore:

$$\hat{y} = \theta_1 + 2\theta_2$$

The squared error function is:

$$f(\theta) = (\hat{y} - y)^2$$

Substituting the prediction and actual value:

$$\boxed{f(\theta_1, \theta_2) = (\theta_1 + 2\theta_2 - 3)^2}$$

1(b)(i) Calculate Partial Derivatives and Gradient

We start with the given error function:

$$f(\theta_1, \theta_2) = (\theta_1 + 2\theta_2 - 3)^2$$

To calculate the derivatives, we use the **Chain Rule**.

The chain rule states that:

$$\frac{d}{d\theta}[g(\theta)]^2 = 2g(\theta) \cdot g'(\theta)$$

For simplicity, let:

$$g(\theta) = \theta_1 + 2\theta_2 - 3$$

Therefore:

$$f(\theta) = g(\theta)^2$$

Step 1: Partial Derivative with Respect to θ_1

Using the chain rule:

$$\frac{\partial f}{\partial \theta_1} = 2(\theta_1 + 2\theta_2 - 3) \cdot \frac{\partial}{\partial \theta_1}(\theta_1 + 2\theta_2 - 3)$$

Since:

$$\frac{\partial \theta_1}{\partial \theta_1} = 1$$

and θ_2 is treated as a constant:

$$\frac{\partial f}{\partial \theta_1} = 2(\theta_1 + 2\theta_2 - 3) \times 1$$

Therefore:

$$\boxed{\frac{\partial f}{\partial \theta_1} = 2(\theta_1 + 2\theta_2 - 3)}$$

Step 2: Partial Derivative with Respect to θ_2

Again, using the chain rule:

$$\frac{\partial f}{\partial \theta_2} = 2(\theta_1 + 2\theta_2 - 3) \cdot \frac{\partial}{\partial \theta_2}(\theta_1 + 2\theta_2 - 3)$$

Since:

$$\frac{\partial (2\theta_2)}{\partial \theta_2} = 2$$

we get:

$$\frac{\partial f}{\partial \theta_2} = 2(\theta_1 + 2\theta_2 - 3) \times 2$$

Therefore:

$$\boxed{\frac{\partial f}{\partial \theta_2} = 4(\theta_1 + 2\theta_2 - 3)}$$

Step 3: Calculate the Gradient Vector

Combining both partial derivatives:

$$\boxed{\nabla_\theta f = \begin{bmatrix} 2(\theta_1 + 2\theta_2 - 3) \\ 4(\theta_1 + 2\theta_2 - 3) \end{bmatrix}}$$

Step 4: Evaluate at the Given Parameters

The question gives:

$$\theta = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Therefore:

$$\theta_1 = 1, \qquad \theta_2 = 0$$

First, calculate the error:

$$f(1,0) = (1 + 2(0) - 3)^2$$

$$= (-2)^2$$

$$\boxed{f(1,0) = 4}$$

Now calculate the first partial derivative:

$$\frac{\partial f}{\partial \theta_1} = 2(1 + 2(0) - 3)$$

$$= 2(-2)$$

$$\boxed{\frac{\partial f}{\partial \theta_1} = -4}$$

Calculate the second partial derivative:

$$\frac{\partial f}{\partial \theta_2} = 4(1 + 2(0) - 3)$$

$$= 4(-2)$$

$$\boxed{\frac{\partial f}{\partial \theta_2} = -8}$$

Therefore, the final gradient is:

$$\boxed{\nabla_\theta f(1,0) = \begin{bmatrix} -4 \\ -8 \end{bmatrix}}$$

Final Answer for 1(b)(i)

$$\boxed{f(1,0) = 4}$$

$$\boxed{\nabla_\theta f(1,0) = \begin{bmatrix} -4 \\ -8 \end{bmatrix}}$$

1(b)(ii) Perform One Gradient Descent Step for Each Learning Rate

We need to perform one gradient descent update for three different learning rates:

$$\eta \in \{0.1,\ 0.2,\ 0.3\}$$

The gradient descent formula is:

$$\boxed{\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla_\theta f}$$

We already know:

$$\theta_{\text{old}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

And:

$$\nabla_\theta f = \begin{bmatrix} -4 \\ -8 \end{bmatrix}$$

We now calculate each case separately.

Case 1: Learning Rate $\eta = 0.1$

Substitute the values into the gradient descent formula:

$$\theta_{\text{new}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} -4 \\ -8 \end{bmatrix}$$

First, multiply the gradient by the learning rate:

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} -0.4 \\ -0.8 \end{bmatrix}$$

Subtract the vectors:

$$\boxed{\theta_{\text{new}} = \begin{bmatrix} 1.4 \\ 0.8 \end{bmatrix}}$$

Now calculate the new error:

$$f(\theta_1, \theta_2) = (\theta_1 + 2\theta_2 - 3)^2$$

Substitute the updated parameters:

$$f(1.4, 0.8) = (1.4 + 2(0.8) - 3)^2$$

$$= (1.4 + 1.6 - 3)^2$$

$$= (0)^2$$

Therefore:

$$\boxed{f_{\text{new}} = 0}$$

The error decreased from 4 to 0.

Case 2: Learning Rate $\eta = 0.2$

Using the same formula:

$$\theta_{\text{new}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 0.2 \begin{bmatrix} -4 \\ -8 \end{bmatrix}$$

Multiply:

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} -0.8 \\ -1.6 \end{bmatrix}$$

Subtract:

$$\boxed{\theta_{\text{new}} = \begin{bmatrix} 1.8 \\ 1.6 \end{bmatrix}}$$

Calculate the new error:

$$f(1.8, 1.6) = (1.8 + 2(1.6) - 3)^2$$

$$= (1.8 + 3.2 - 3)^2$$

$$= (2)^2$$

Therefore:

$$\boxed{f_{\text{new}} = 4}$$

The error remains unchanged at 4.

Case 3: Learning Rate $\eta = 0.3$

Again, using gradient descent:

$$\theta_{\text{new}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 0.3 \begin{bmatrix} -4 \\ -8 \end{bmatrix}$$

Multiply:

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} -1.2 \\ -2.4 \end{bmatrix}$$

Subtract:

$$\boxed{\theta_{\text{new}} = \begin{bmatrix} 2.2 \\ 2.4 \end{bmatrix}}$$

Calculate the new error:

$$f(2.2, 2.4) = (2.2 + 2(2.4) - 3)^2$$

$$= (2.2 + 4.8 - 3)^2$$

$$= (4)^2$$

Therefore:

$$\boxed{f_{\text{new}} = 16}$$

The error increased from 4 to 16.

Final Comparison

Learning Rate	Updated Parameters	New Error
$\eta = 0.1$	$(1.4,\ 0.8)$	0
$\eta = 0.2$	$(1.8,\ 1.6)$	4
$\eta = 0.3$	$(2.2,\ 2.4)$	16

Observation and Conclusion

The learning rate determines the size of the step taken during gradient descent.

- **Learning rate 0.1:** The updated parameters reach the minimum of the error function in one step. The error decreases from 4 to 0.
- **Learning rate 0.2:** The update overshoots the minimum. The new error is 4, which is the same as the original error.
- **Learning rate 0.3:** The update overshoots the minimum even further. The new error increases to 16.

This shows that choosing the correct learning rate is very important.

A learning rate that is too small can make learning slow, while a learning rate that is too large can cause the algorithm to overshoot the minimum and increase the error.